Information System Management

Course No: 0612-301

Introduction to Information Systems

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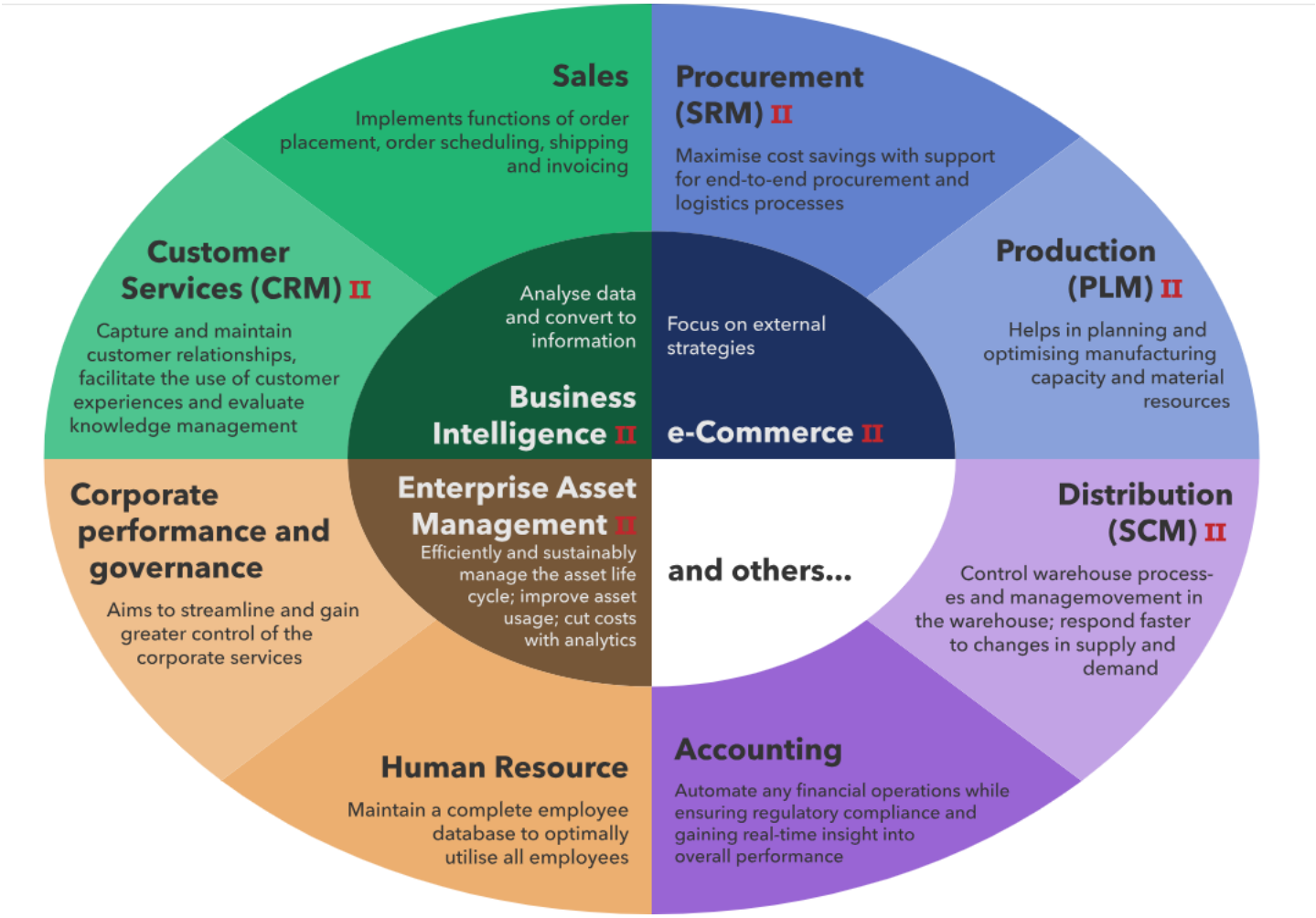
Lecturer, CSE

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System Concept

- A system –
  - A way of thinking about organizations and their problems.
  - Involves a set of techniques that helps in solving problems.
  - An organized relationship among functioning units.
  - An orderly grouping of individual components linked together to achieve specific objective/s.
  - Each component is part of the total system and has to do its share of work for the system to achieve the intended goal.
  - Components may refer to physical parts, managerial steps, etc.

System Concept



System Concept

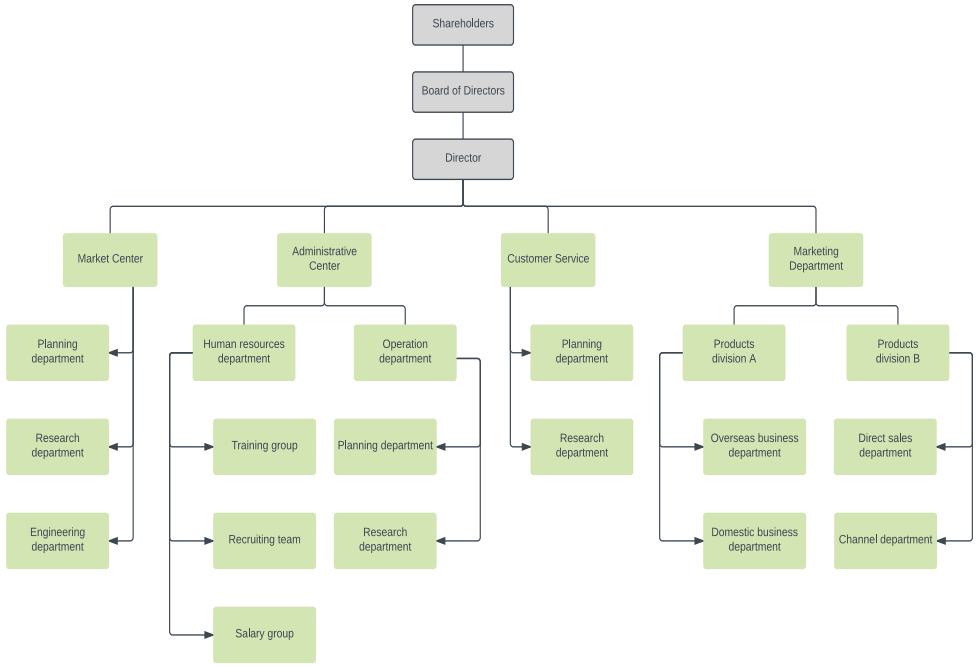
- Therefore, The study of systems concepts has three basic implications –
  - A system must be designed to achieve a predetermined objective
  - Interrelationships and interdependence must exist among the components.
  - The objectives of the organization as a whole have a higher priority than the objectives of its subsystems.

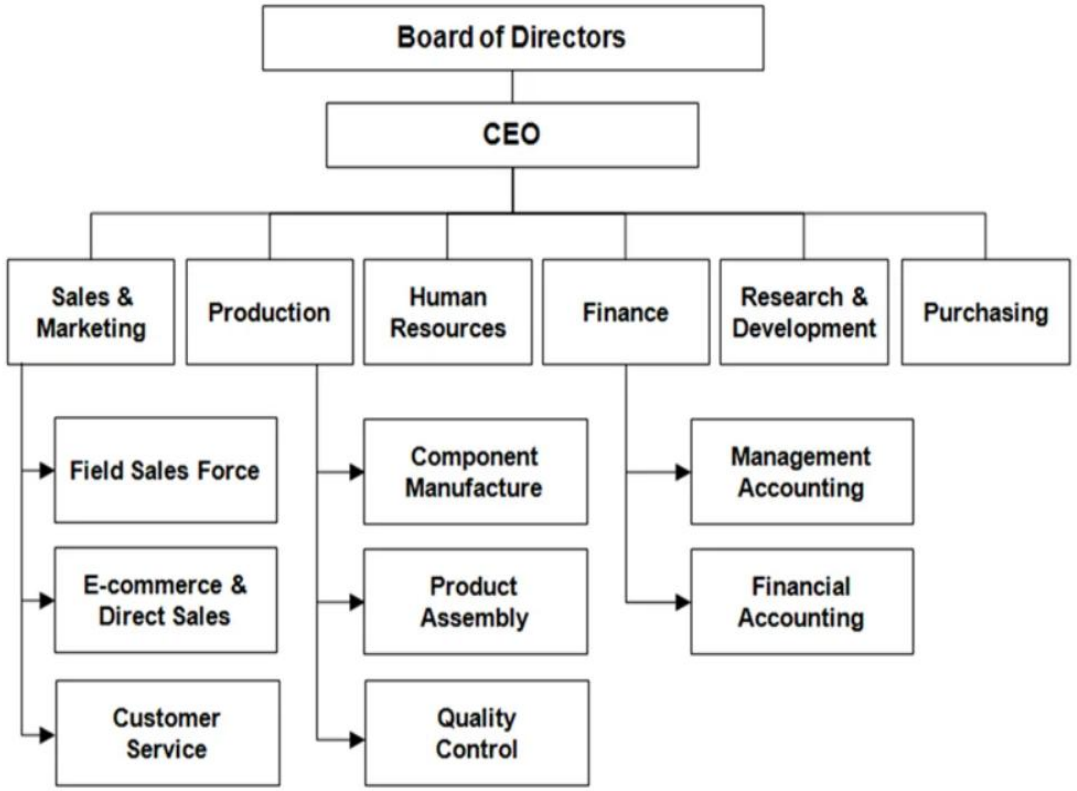
Characteristics of a System

- Characteristics of a system
  - Organization
  - Interaction
  - Interdependence
  - Integrations
  - Central objective

Characteristics of a System

- Organization
  - Implies structure and order.
  - The arrangement of components that helps to achieve objectives.





Characteristics of a System

- Interaction
  - How each component functions with other components.
  - Communication and exchange of information, energy, or resources between the components of a system.
  - Essential for coordination and functionality.

- Interdependence
  - Dependence of each subsystem or component on others.
  - Certain subsystem may need the outputs of other subsystem as input.
  - Ensures the contribution of each subsystem to achieve the system objectives.

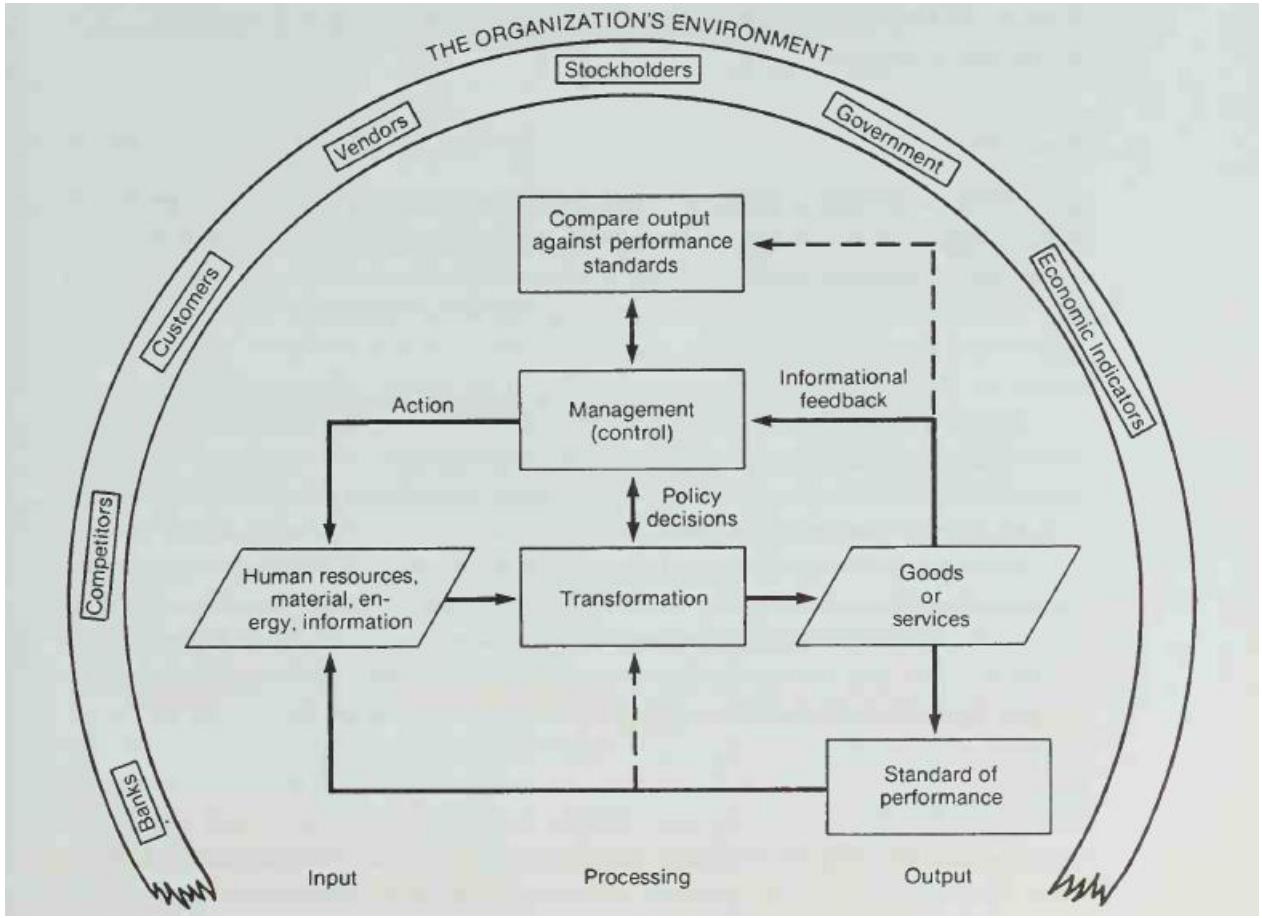
Characteristics of a System

- Integration
  - Concerned with how a system is tied together.
  - The harmonious functioning of all components as a unified whole.
  - All the parts of the system need to work in sync even if their jobs are unique.
  - Proper integration maximizes the system's output and effectiveness.

- Central Objective
  - Guides the functioning and alignment of all components.
  - It can be real or stated.
    - A company might state that its primary objective is to "serve customers," but in reality, its operations might prioritize maximizing profit.

Elements of a System

- Inputs and outputs
- Processors
- Control
- Feedback
- Environment
- Boundaries and interface



Elements of a System

- Inputs and outputs
  - A system's main objective is to produce an output that meets the expectations of its intended users.
  - Inputs are processed by the system to generate the desired output.
  - Determining the output is the first step in defining the required input and processing design, as seen in systems analysis.

- Processors
  - Involves the actual transformation of input into output.
  - It is the operational component of a system.

Elements of a System

- Control
  - A decision-making subsystem.
  - Governs the input, processing, and output of a system to ensure its balance.

- Feedback
  - Control in a dynamic system is achieved through feedback.
  - Measures output against standards and may lead to changes in input, processing, or output.
  - Can be positive or negative.
  - Vital to identify issues, justify changes, and improve performance.

Elements of a System

- Environment
  - The external elements that impacts the system.
  - Often determines how a system must function.

- Boundaries and Interface
  - A system’s boundaries determines its scope and interface with other systems.
  - Boundaries help identify what a system includes and excludes.

Types of Systems

- Physical systems
  - Static or dynamic
- Abstract systems
  - Conceptual models to point out the significant components and relationships.
- Open and closed systems
  - Based on the system's degree of independence.
- Information systems

Open Systems

- Has many interfaces with its environment.
- Permits interaction across its boundary to exchange inputs and outputs.
- Five important characteristics –
  - Input from outside
    - Self-adjusting and self-regulating to achieve steady state i.e. equilibrium.
  - Entropy
    - To get to steady state and not to run down over time.
  - Process, output, and cycles
    - Follows a continuous flow path.
  - Differentiation
    - Greater specialization and interaction to increase value.
  - Equifinality
    - Goals are achieved through a variety of paths.

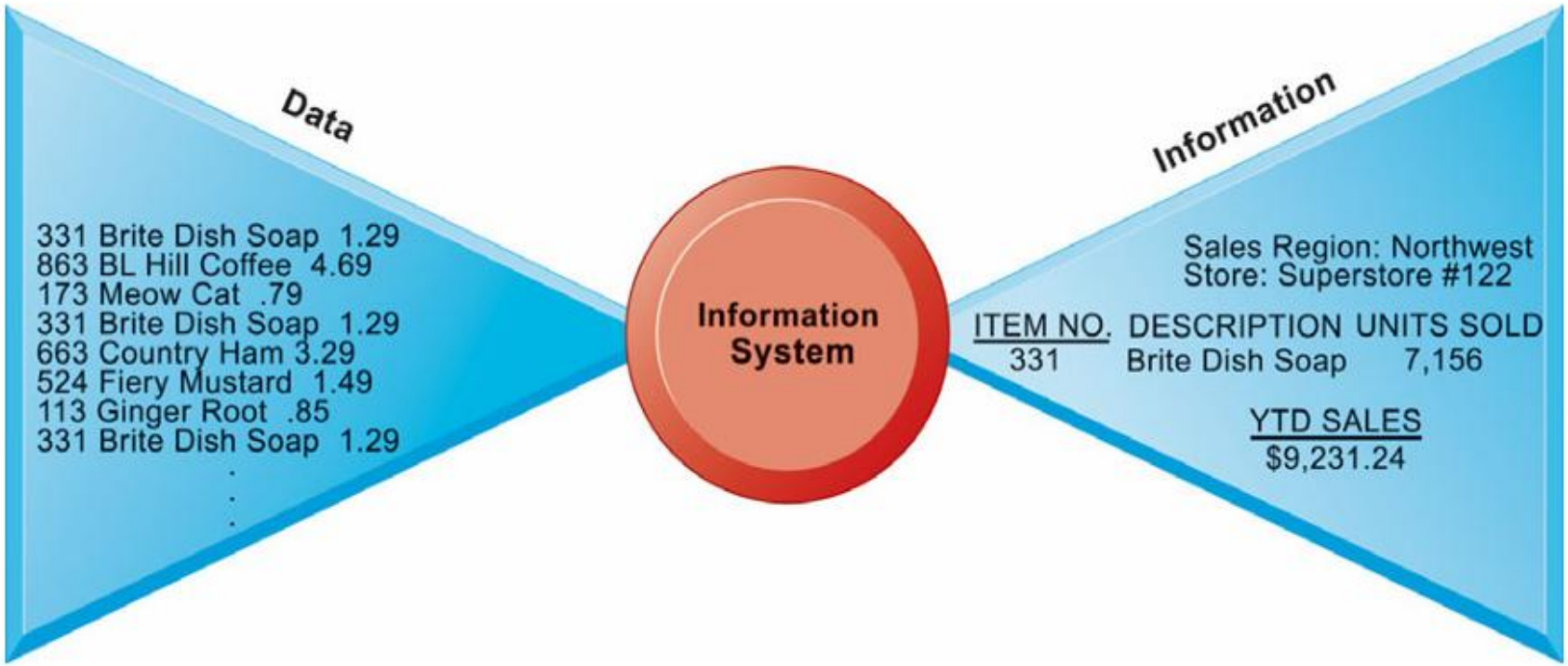
Information Systems

- Information system –
  - a set of interrelated components
  - that collect (or retrieve), process, store, and distribute information
  - to support decision making and control in an organization

- Data are streams of raw facts representing events in the organizations and the environment.

- Information means data that have been shaped into a form that is meaningful and useful to human beings.

Information Systems

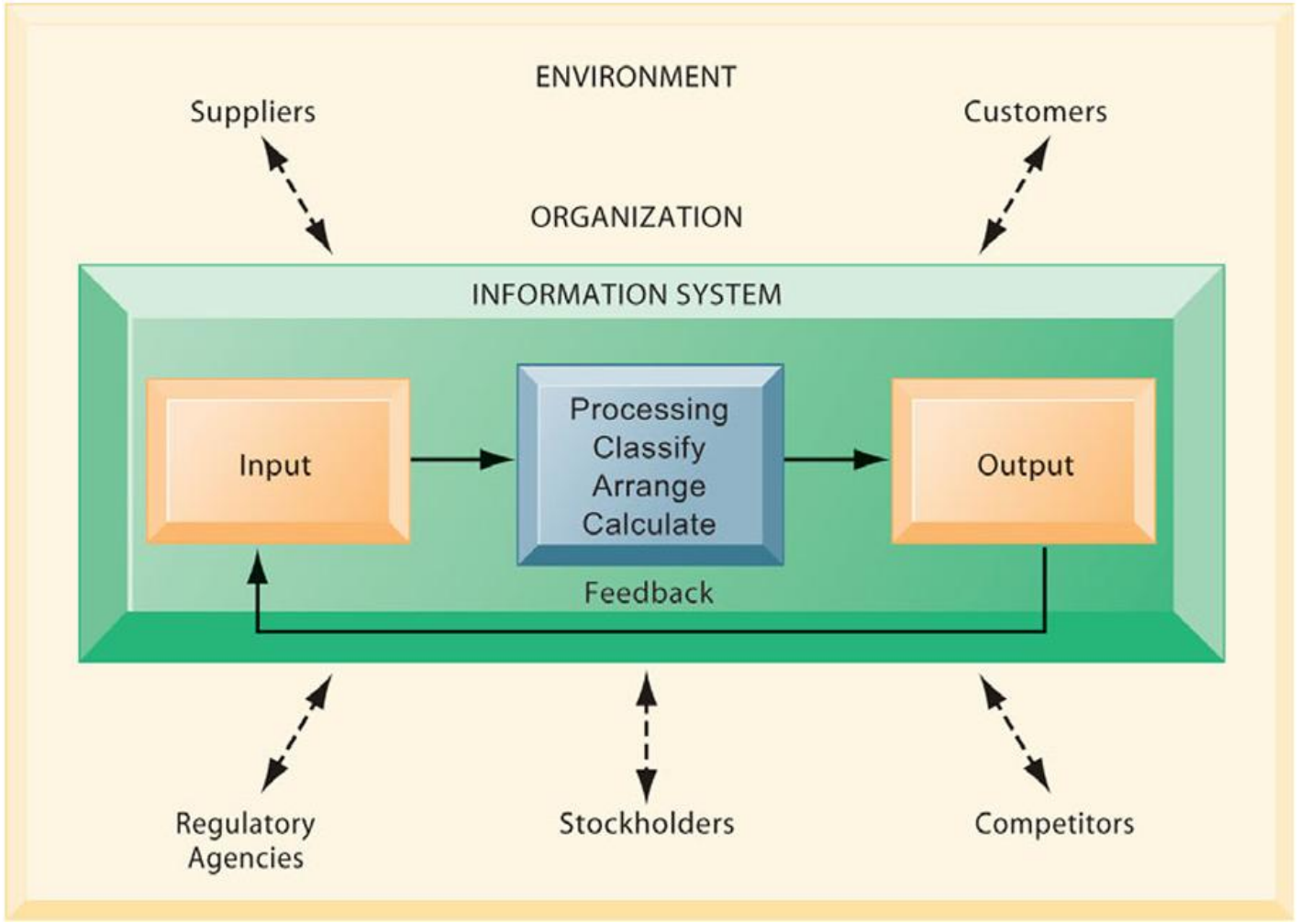


Raw data of a supermarket are being processed to extract useful information

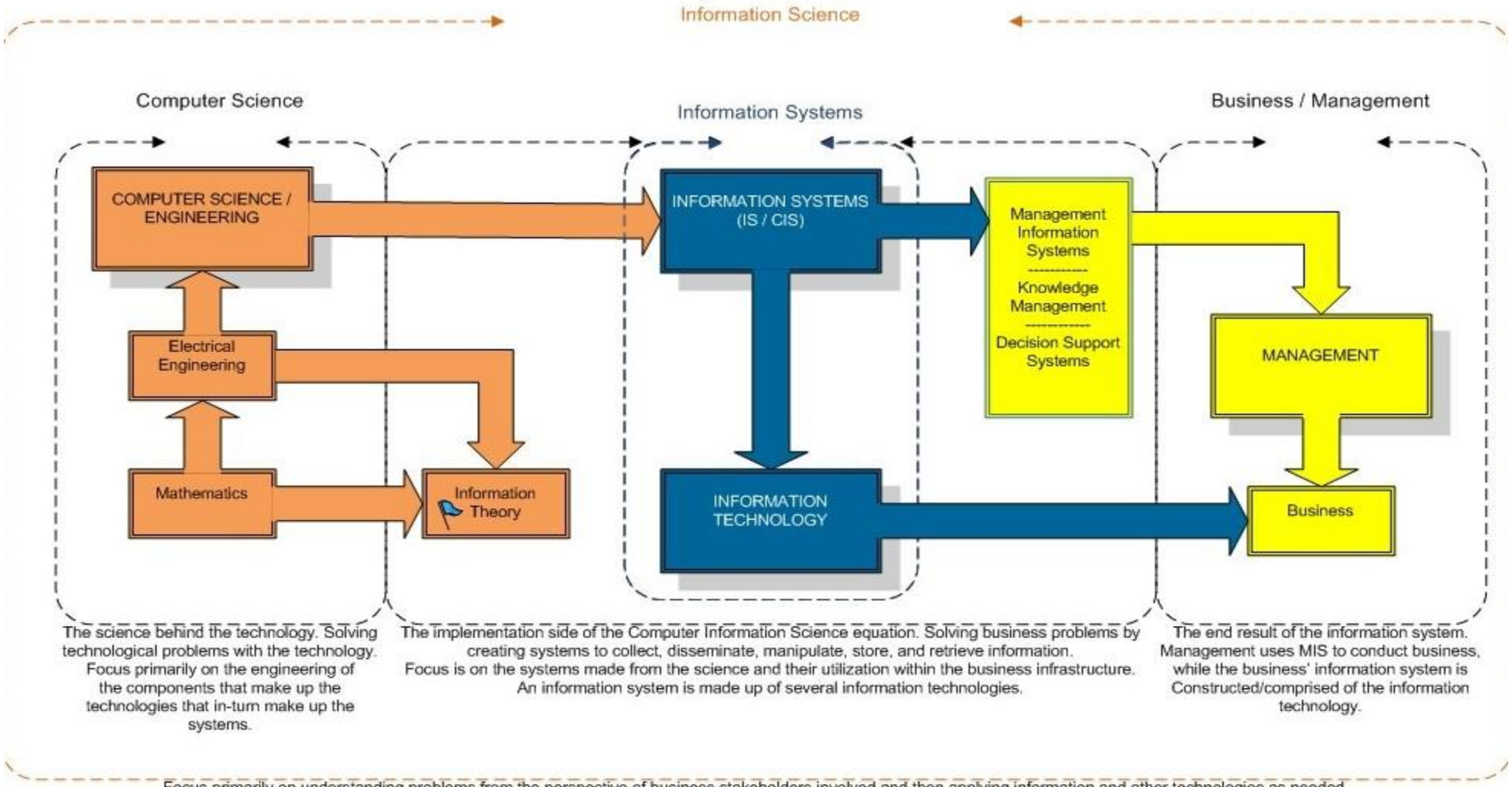
Information Systems

- The activities/functions/elements of information systems –
  - Environment
    - The external factors that impact and shape the system
  - Inputs
    - Captures or collects raw data from within the organization or from its external environment
  - Processing
    - Converts this raw input into a meaningful form
  - Outputs
    - Transfers the processed information to wherever it will be used
  - Feedback
    - Output that is returned to appropriate members of the organization to help them evaluate or correct the input stage

Information Systems



Information Systems

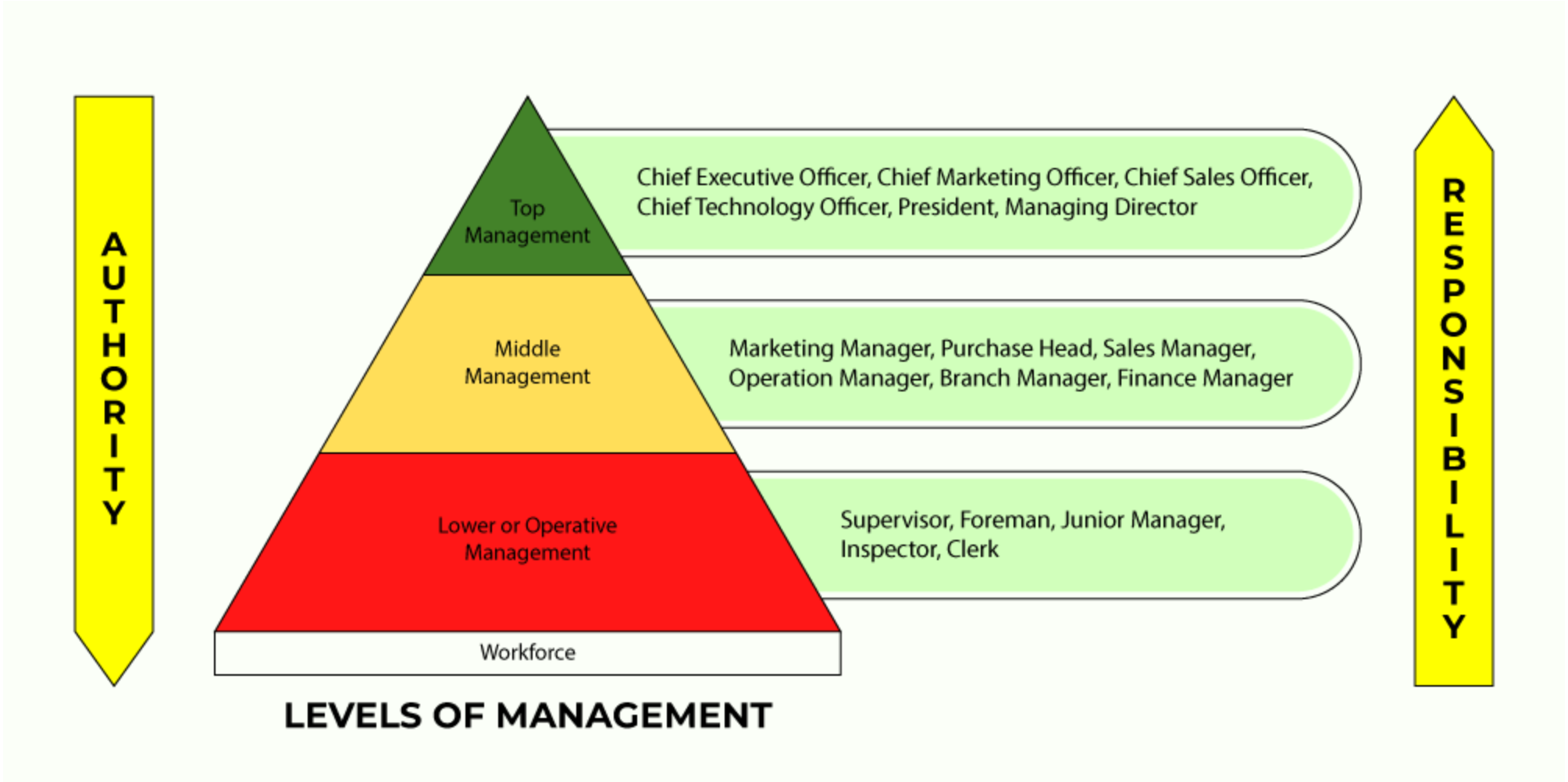


Dimensions of Information Systems



- Computers and softwares are not the only part of an information system.
- The dimensions of information system –
  - Organizations
  - Management
  - Information technology
- They address –
  - The problems they are designed to solve
  - Their architectural and design elements
  - The organizational processes that lead to these solutions

Organizations



Organizations

- Top level management
  - Formulates objectives, plans, and policies.
  - Makes long-range strategic decisions.
  - Coordinates and control of the performance.
  - Analyses of the business environment.
  - Sets up an organizational framework.
  - Assembles of the resources.
  - Ensures financial performance of the firm.
  - Requires summarized information from a variety of sources to attain goals.
  - Relatively open, dynamic, and adaptive.
  - Decisions are mostly judgmental.

Organizations

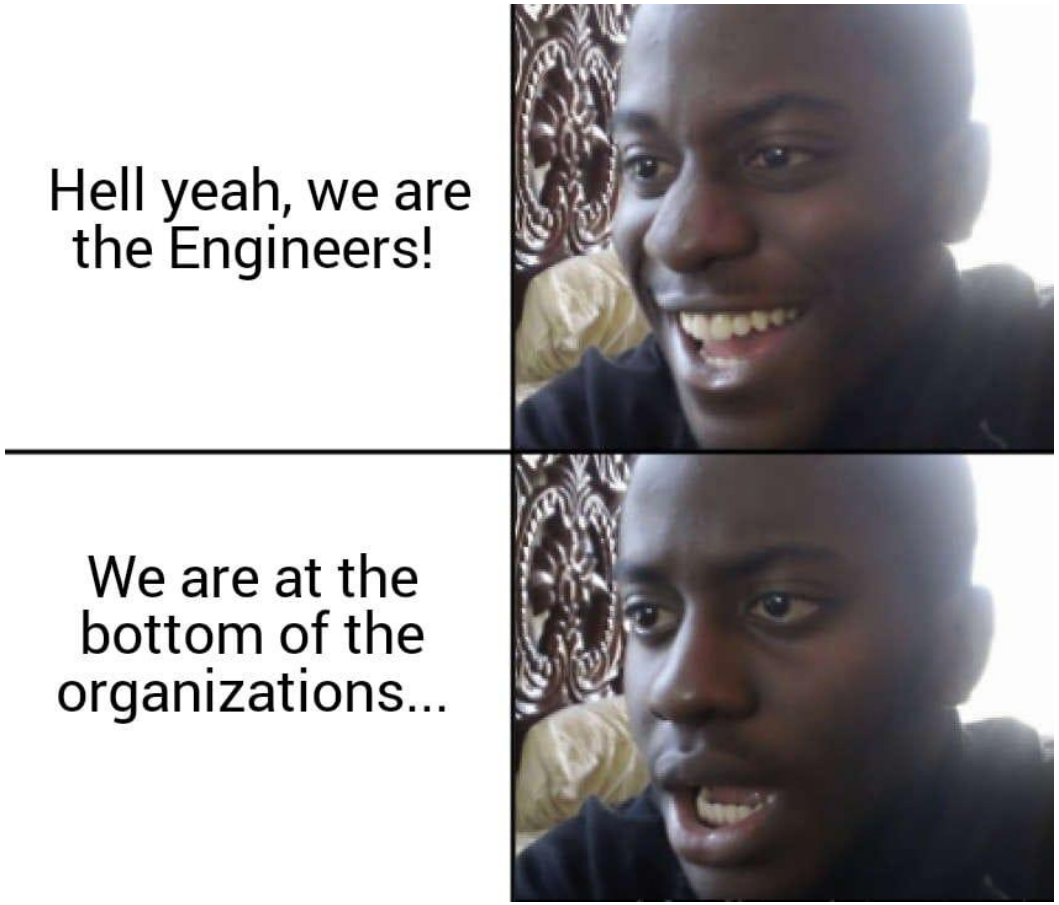
- Middle management
  - Carries out the programs and plans of senior management.
  - Interprets of the policies framed by the Top Level Management.
  - Selects of suitable operative and supervisory personnel.
  - Assigns of duties and responsibilities to the Lower Level Management.
  - Motivates employees to get desired objectives.
  - Cooperates with the entire organization.

Organizations

- Lower management
  - Also known as Supervisory or Operational Level Management.
  - Issues of orders and instructions.
  - Makes day-to-day, relatively structured control decisions.
  - Monitors the daily activities of the business.
  - Assigns and assists the workers
  - Represents workers' grievances.
  - Ensures a safe and proper work environment.
  - Helps the middle level management.
  - Encourages initiative of employees.
  - Needs detailed internal information.

Organizations

- Workforces include –
  - Data workers
    - Clerks, secretariats, etc.
  - Knowledge workers
    - Engineers (😔), scientists, architects, etc.
  - Production workers
  - Service workers
  - And, many others



Organizations

- Business functions
  - Sales, marketing, manufacturing, production, finance, accounting, HR, etc.
- Formal rules
- Informal work practices
- Automation
- Differing perspectives, conflicts, compromises, and agreements.
- Unique culture
  - Fundamental set of assumptions, values, and ways of doing things.
  - Accepted by most of its members.

Dimensions of Information Systems

- Management
  - Manage what already exists.
  - Create new products and services.
  - If needed, re-create the organization.

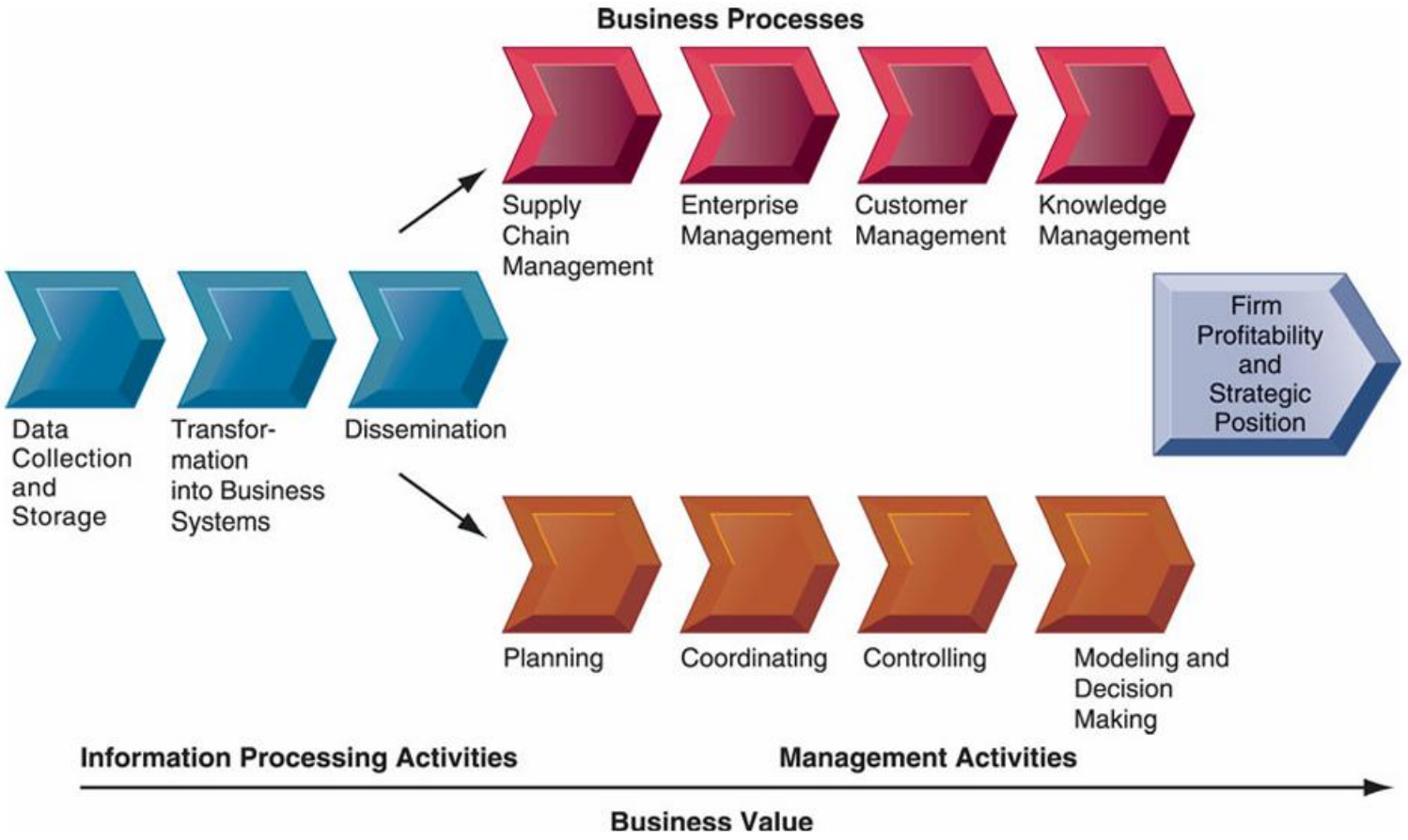
- Information Technology Infrastructures
  - Provides the foundation, or platform, on which the firm can build its IS.
  - Hardware and software.
  - Data management technology.
  - Networking and telecommunications technology.

Information Systems

- Information system enables the firm to –
  - increase its revenue or decrease its costs
  - by providing information
  - that helps managers make better decisions
  - that improves the execution of business processes

- The decision to invest in any new information system is determined by the extent to which the system will lead to –
  - better management decisions
  - more efficient business processes
  - higher firm profitability

Information Value Chain



Complementary Assets

- Investing in information technology does not by itself guarantee good returns.
- They must be accompanied by complementary assets.
- Complementary assets are those assets required to derive value from a primary investment.
- They are the supportive values, structures, behavior patterns, etc. in the organization.

Complementary Assets

Organizational assets	Supportive organizational culture that values efficiency and effectiveness Appropriate business model Efficient business processes Decentralized authority Distributed decision-making rights Strong IS development team
Managerial assets	Strong senior management support for technology investment and change Incentives for management innovation Teamwork and collaborative work environments Training programs to enhance management decision skills Management culture that values flexibility and knowledge-based decision making.
Social assets	The Internet and telecommunications infrastructure IT-enriched educational programs raising labor force computer literacy Standards (both government and private sector) Laws and regulations creating fair, stable market environments Technology and service firms in adjacent markets to assist implementation

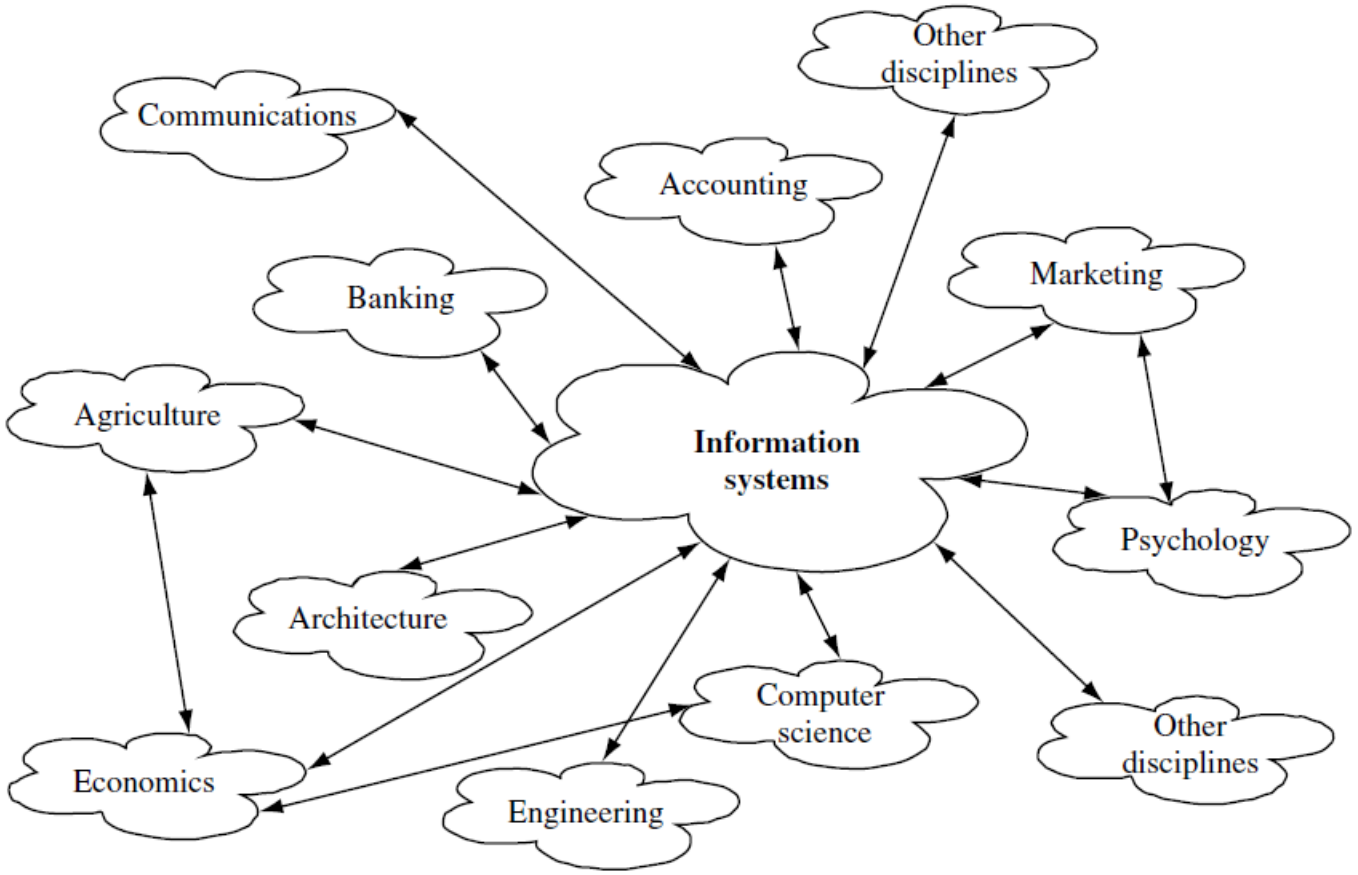
Study on Information Systems

- The study of information systems and their development is a multi disciplinary subject.

- It addresses the range of strategic, managerial and operational activities involved in the gathering, processing, storing, distributing and use of information, and its associated technologies, in society and organizations.

- It examines more than just the technological system, or just the social system. It investigates the phenomena that emerge when the two interact.

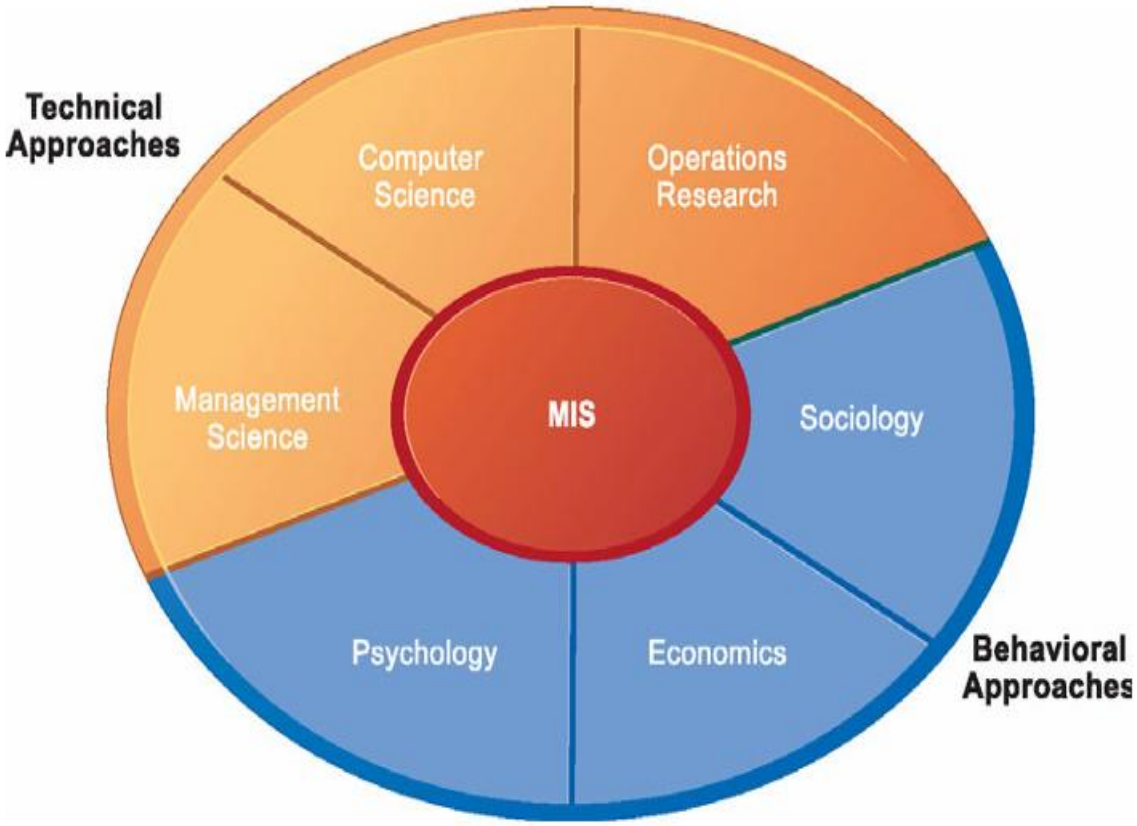
Study on Information Systems



Study on Information Systems

- Also, there are many categories of information systems –
  - Artificial intelligence system
  - Computing platform
  - Data warehouses
  - Decision support system
  - Enterprise resource planning
  - Enterprise systems
  - Expert systems
  - Global information system
  - Multimedia information system
  - Office automation
  - Process control system
  - Search engines
  - Social information systems
  - Geographic information system

Approaches to Study Information Systems



- This field is called Information Management System.
- Technical approaches
- Behavioral approaches
- Socio-technical approaches

Approaches to Study Information Systems

- Technical approaches –
  - Emphasizes mathematically based models to study information systems.
  - Physical technology and the formal capabilities of these systems.

- Behavioral approaches –
  - Strategic business integration, design, implementation, utilization, and management cannot be explored usefully with the models used in the technical approach.
  - Concentrates on changes in attitudes, management and organizational policy, and behavior.

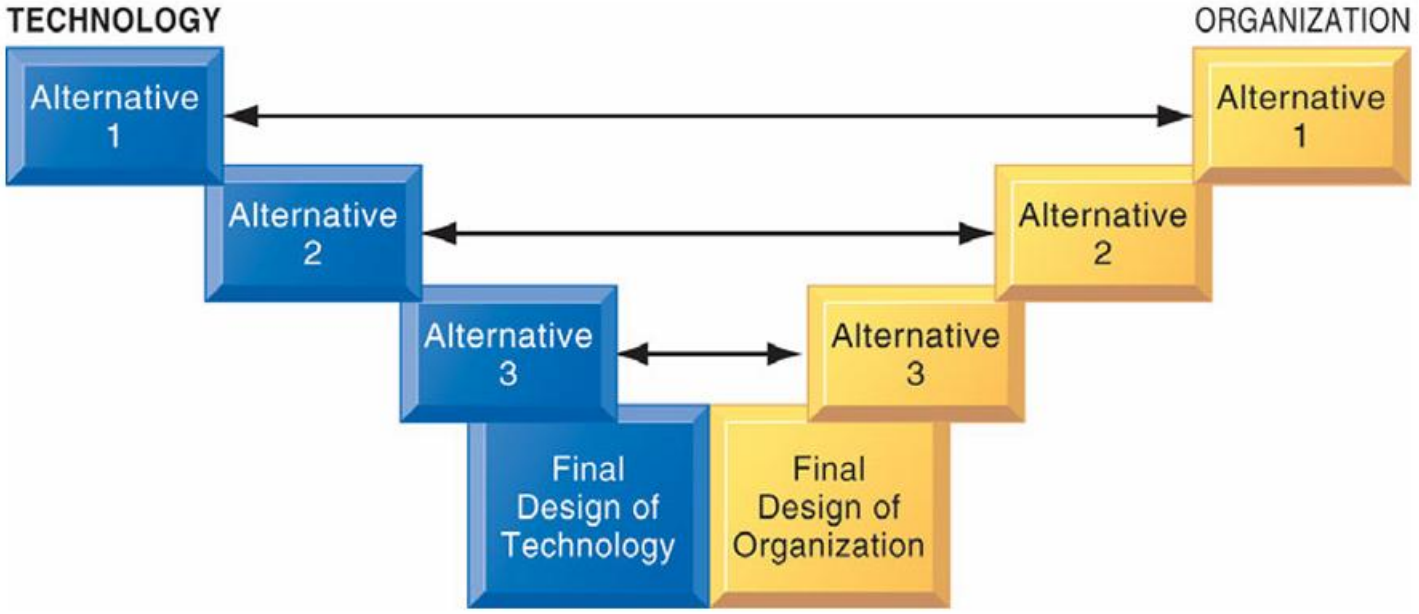
Socio-technical Approaches

- Information systems management combines –
  - the work of computer science, management science, and operations research
  - with a practical orientation
  - toward developing system solutions to real-world problems
  - and managing information technology resources.

- It is also concerned with –
  - behavioral issues surrounding the development, usage, and impact of information systems
  - which are typically discussed in the fields of sociology, economics, and psychology

Socio-technical Approaches

- In this view, optimal organizational performance is achieved by jointly optimizing both the social and technical systems.
- Technology must be changed and designed in such a way as to fit organizational and individual needs.



References

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The END